914528

Course Name: DATABASE MANAGEMENT SYSTEMS

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data model used in database design.
CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real-life application, with constraints and keys using SQL.
CO5: Select the information from a database by formulating complex queries in SQL.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

Printed Pages: 03

University Roll No. .....

Mid Term Examination, Odd Semester 2025-26
Program: BTech (CSE), Year: II, Semester: III
Subject Code: BCSC 1003
Subject Name: Database Management Systems

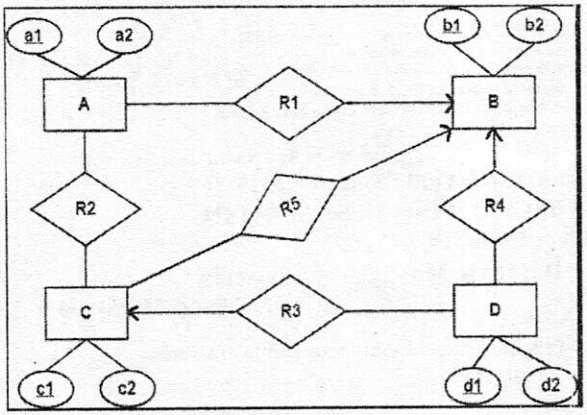
Time: 2 Hours **Maximum Marks: 30**

Instruction for students: Attempt both sections and use only blue pen to answer.

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Differentiate between a traditional File System and a Database Management System (DBMS). Highlight the limitations of file systems in terms of data management, and explain how a DBMS overcomes those limitations. Support your answer with a practical example scenario where using a DBMS is more efficient and reliable compared to a file system.	3	1	R	C
2	In the context of the ANSI–SPARC three-level architecture, define Data Independence. Explain the two forms of data independence: logical data independence, and physical data independence. Support your explanation with suitable examples to illustrate how DBMS provides abstraction and schema modification transparency.	3	1	A	F
3	Design a database schema with at least 3 relations that collectively enforce following integrity constraints: Entity, Domain, and Referential.	3	4	A	C
4	Define anomalies in the context of functional dependencies and unnormalized relations. Also explain 3NF in details along with a suitable example.	3	6	U	F

5	Find the minimum numbers of table required to represent the given ER diagram into relational model:  	3	2	R	F

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider the following relational schemas: STUDENT (RollNo, StdName, City) TEACHER (TeacherID, TeacherName, Subject) ENROLLMENT (RollNo, TeacherID) Write relational algebra expressions for the following queries: a. Find the roll number and name of all students. b. Find the name of teacher who teaches DBMS. c. Find the name of teacher who teaches Machine Learning to Sachin. d. Insert a new tuple into Student relation (Assume record of your own choice). e. Delete records of students whose location is Gurugram.	5	5	C	P

7	Consider a relation schema R (W, X, Y, Z) with the following set of functional dependencies (FDs): $\{WX \rightarrow YZ, WY \rightarrow XZ, XY \rightarrow Z\}$ Determine: • The Candidate Key(s) of the relation. • The Prime Attribute(s) & Non-Prime Attribute(s). • The highest normal form that the relation satisfies.	5	6	E	P
8	The Indian Premier League (IPL) wants to maintain a database for managing the tournament. The IPL has several Teams. Each team has a unique team ID, name, city, and owner. Each Team consists of many Players. A player has a player ID, name, role (batsman, bowler, all-rounder, wicket-keeper), nationality, and age. Each Match is played between two teams on a particular date, at a stadium. The match has a unique match ID, venue, and result. For every match, Player Performance is recorded (runs scored, wickets taken, catches). Umpires officiate matches. Each umpire has an umpire ID, name, and nationality. A match can have multiple umpires. Sponsors are associated with teams. A sponsor has sponsor ID, name, and deal amount. One sponsor can sponsor multiple teams. Draw an ER diagram for this IPL database.	5	3	C	P